PSEUDOCOMPLETENESS OF COUNTABLE CLOSURES IN SPACES OF CONTINUOUS FUNCTIONS

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Abstract. We answer Question 4.8 of Dow and Tkachuk affirmatively in ZFC: if the closure of every countable subset of $C_p(X)$ is pseudocomplete, then the Tychonoff space X is a P-space. More precisely, every non-P-space X admits a closed separable metrizable linear subspace of $C_p(X)$ that is not pseudocomplete. The proof interpolates sequences at levels tending to zero: finitely supported sequences give continuous functions, whereas the constant sequence yields the discontinuous characteristic function of the complement of a non-open zero set. Coordinate evaluation helps make the interpolation image closed, and the Aarts–Lutzer characterization supplies the contradiction.

1. The result and the argument

All spaces considered below are Tychonoff. We give $C_p(X) = C(X, \mathbb{R})$ the topology inherited from $\mathbb{R}^X$, and write $\mathbb{N} = \{1, 2, \ldots\}$. A *$P$-space* is a space in which every countable intersection of open sets is open.

Dow and Tkachuk [2, Question 4.8] ask whether pseudocompleteness of the closure of every countable subset of $C_p(X)$ forces X to be a P-space. Their Theorem 3.14 yields pseudocompleteness of $C_p(X)$ from this hypothesis; the conclusion about X requires an additional argument. We prove the following stronger obstruction statement.

Theorem 1.1. *If X is not a P-space, then $C_p(X)$ contains a closed separable metrizable linear subspace which is not pseudocomplete.*

Corollary 1.2. *If $\overline{A}^{C_p(X)}$ is pseudocomplete for every countable $A \subseteq C_p(X)$, then X is a P-space.*

The key idea is to encode a sequence a by a function satisfying $T(a)(x_n) = a_n$, using linear interpolation at levels $t_n \to 0$. Finite support makes the interpolant continuous at zero. In contrast, the constant sequence gives the characteristic function of $X \setminus Z$, which is discontinuous. Coordinate evaluation provides a continuous left inverse, and the interpolation image is closed. Pseudocompleteness would force the constant sequence into the same domain as the finitely supported sequences, yielding the contradiction.

2. The completeness obstruction

We use pseudocompleteness in Oxtoby's sense [4], as in [1, Section 2.1] and [2, Section 2]. A sequence $(\mathcal{B}_n)$ of π-bases witnesses pseudocompleteness if, whenever $U_n \in \mathcal{B}_n$ and $\overline{U_{n+1}} \subseteq U_n$ for every n, the intersection $\bigcap_{n\geq 1} U_n$ is nonempty. Here a π-base consists of nonempty open sets, and every nonempty open set contains one of its members.

Lemma 2.1. *A dense pseudocomplete additive subgroup of $\mathbb{R}^{\mathbb{N}}$ is equal to $\mathbb{R}^{\mathbb{N}}$.*

Proof. Let G be such a subgroup. By [1, Section 2.4, Corollary, p. 4], a metrizable pseudocomplete space contains a dense completely metrizable subspace; apply this to G and take $D \subseteq G$ with this property. Since D is dense in G and G is dense in $\mathbb{R}^{\mathbb{N}}$, D is dense in $\mathbb{R}^{\mathbb{N}}$ as well. A completely metrizable subspace of a metrizable space is a G_δ subset of that space [3, Theorem 4.3.24]. Thus D is a dense G_δ in the ambient product $\mathbb{R}^{\mathbb{N}}$, not merely in G. (For metrizable spaces, complete metrizability is equivalent to Čech-completeness [3, Theorem 4.3.26], so this agrees with the formulation of the characterization in [2, Section 2].)

Fix an arbitrary $a \in \mathbb{R}^{\mathbb{N}}$. Translation is a homeomorphism of $\mathbb{R}^{\mathbb{N}}$, so $a + D$ is another dense G_δ subset. Since $\mathbb{R}^{\mathbb{N}}$ is completely metrizable, the Baire category theorem gives $D \cap (a + D) \neq \emptyset$. Hence $d_1 = a + d_2$ for some $d_1, d_2 \in D$, and $a = d_1 - d_2 \in G$. Since a was arbitrary, $G = \mathbb{R}^{\mathbb{N}}$. □

The same proof applies to a dense pseudocomplete subgroup of any completely metrizable topological group. Only the additive version above is needed here.

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3. The closed interpolation subspace

Proof of Theorem 1.1. A Tychonoff space is a P-space precisely when all its zero sets are open; a proof is recalled in Section 4. Choose a continuous $f\colon X \to [0,1]$ such that $Z = f^{-1}(\{0\})$ is not open. Positive values in $f(X)$ occur arbitrarily close to zero. Otherwise $Z = f^{-1}([0,\varepsilon))$ for some $\varepsilon > 0$, which would make Z open.

Choose $x_n \in X$ so that, with $t_n = f(x_n)$,

$$0 < t_{n+1} < \frac{t_n}{2} \qquad (n \geq 1).$$

Then $t_n \to 0$ and the x_n are distinct. No convergence of (x_n) in X is needed.

For $a = (a_n)_{n\in\mathbb{N}} \in \mathbb{R}^{\mathbb{N}}$, we construct $u_a\colon [0,1] \to \mathbb{R}$ by linear interpolation between the prescribed values $u_a(t_n) = a_n$, setting $u_a(0) = 0$:

$$u_a(t) = \begin{cases} 0, & t = 0, \\ a_1, & t_1 \leq t \leq 1, \\ \dfrac{t - t_{n+1}}{t_n - t_{n+1}}\, a_n + \dfrac{t_n - t}{t_n - t_{n+1}}\, a_{n+1}, & t_{n+1} \leq t \leq t_n \quad (n \geq 1). \end{cases}$$

The pieces agree at their common endpoints and cover $(0,1]$; moreover, $u_a(t_n) = a_n$ for every n. The function u_a is continuous on $(0,1]$.

For finitely supported a, the function u_a vanishes on a neighborhood of zero and is therefore continuous on all of $[0,1]$.

Define

$$T\colon \mathbb{R}^{\mathbb{N}} \to \mathbb{R}^X, \quad T(a) = u_a \circ f, \qquad P\colon \mathbb{R}^X \to \mathbb{R}^{\mathbb{N}}, \quad P(g) = \big(g(x_n)\big)_{n\in\mathbb{N}}.$$

The map T is linear. For each fixed $x \in X$, the x-coordinate of $T(a)$ is either zero or a fixed linear combination of at most two coordinates of a; hence T is continuous for the product topologies. The evaluation map P is continuous, and $P(T(a)) = a$. Thus T is a homeomorphism onto its image, with inverse the restriction of P. Moreover,

$$L := T(\mathbb{R}^{\mathbb{N}}) = \{g \in \mathbb{R}^X : T(P(g)) = g\}$$

is closed in $\mathbb{R}^X$, since it is the equalizer of two continuous maps into a Hausdorff space.

Let $\mathbb{Q}^{(\mathbb{N})}$ denote the set of finitely supported rational sequences, and set $A = T(\mathbb{Q}^{(\mathbb{N})})$. This set is countable, and the finite-support property above gives $A \subseteq C_p(X)$. Since $\mathbb{Q}^{(\mathbb{N})}$ is dense in $\mathbb{R}^{\mathbb{N}}$ and T is a homeomorphism onto the closed subspace L,

$$\overline{A}^{\mathbb{R}^X} = L.$$

Because $C_p(X)$ has the subspace topology inherited from $\mathbb{R}^X$, it follows that

$$F := \overline{A}^{C_p(X)} = L \cap C_p(X).$$

Thus F is closed in $C_p(X)$, separable (since A is countable and dense in F), metrizable, and real-linear.

Put

$$G = \{a \in \mathbb{R}^{\mathbb{N}} : T(a) \in C_p(X)\}.$$

Then $T|_G\colon G \to F$ is a homeomorphism, with inverse $P|_F$. Indeed, $P \circ T$ is the identity on $\mathbb{R}^{\mathbb{N}}$ and $T \circ P$ is the identity on L. The subspace G is an additive subgroup containing $\mathbb{Q}^{(\mathbb{N})}$, so it is dense in $\mathbb{R}^{\mathbb{N}}$. Both G and F are metrizable.

Suppose F were pseudocomplete. Then G would also be pseudocomplete, and the lemma would give $G = \mathbb{R}^{\mathbb{N}}$. In particular, the constant sequence $\mathbf{1} = (1, 1, \ldots)$ would belong to G. By the interpolation construction,

$$T(\mathbf{1})(x) = \begin{cases} 0, & x \in Z, \\ 1, & x \notin Z. \end{cases}$$

Thus continuity of $T(\mathbf{1})$ would imply that

$$Z = T(\mathbf{1})^{-1}\big((-1/2, 1/2)\big)$$

is open, a contradiction. Consequently, F is not pseudocomplete. □

Proof of Corollary 1.2. The set A constructed above is countable and its closure in $C_p(X)$ is F. Hence the hypothesis rules out the obstruction in Theorem 1.1. □

4. Zero sets and scope

We recall the zero-set characterization used in the proof. Suppose every zero set in X is open, and let $x \in \bigcap_{n\geq 1} U_n$, where the U_n are open. Complete regularity supplies continuous functions $h_n \colon X \to [0,1]$ such that $h_n(x) = 0$ and $h_n = 1$ on $X \setminus U_n$. The series

$$h = \sum_{n\geq 1} 2^{-n} h_n$$

converges uniformly and defines a continuous real-valued function. Its terms are nonnegative, and any point outside some U_n has a strictly positive nth term. Therefore

$$x \in h^{-1}(\{0\}) \subseteq \bigcap_{n\geq 1} U_n.$$

By assumption, this zero set is open. Thus every point of $\bigcap_{n\geq 1} U_n$ has an open neighborhood contained in the intersection, so the intersection is open. Conversely, every zero set is a G_δ, proving the criterion. A real-valued zero-set function can be replaced by $|h|/(1+|h|)$ to obtain a function into $[0,1]$ with the same zero set.

Theorem 1.1 shows that whenever X is not a P-space, a failure of pseudocompleteness can already be witnessed by a closed separable metrizable linear subspace of $C_p(X)$. The argument does not use inheritance of pseudocompleteness by arbitrary closed subspaces: in Corollary 1.2 the hypothesis is applied directly to the countable set A. Nor does the argument assume functional countability, or settle Question 4.9 of [2]. All constructions and completeness results used here are in ZFC.

Disclosure of AI assistance

This work is AI-assisted. The main idea is due to the author: to answer Question 4.8 by proving the stronger statement of Theorem 1.1, namely that a non-P-space X always yields a closed separable metrizable linear subspace of $C_p(X)$ that is not pseudocomplete. ChatGPT (Astra, OpenAI) worked out the details and completed the argument. The author edited the manuscript and checked the literature. Claude Opus 5.5 (Anthropic) assisted with reviewing the proof and typesetting, and produced a Lean 4/Mathlib formalization of Corollary 1.2, which compiles without `sorry` and uses only Lean's standard axioms; it is available at `https://github.com/LucasAlves8Sp/dow-tkachuk-question-4-8`. No costs were incurred beyond the author's existing US$20 monthly subscriptions to ChatGPT (OpenAI) and Claude (Anthropic). The result has not yet been checked by a human expert and should be regarded as a candidate resolution of the question.

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